

Homework 3

Due Friday, February 27

1. For a convex set K , let $\mathcal{F}(K)$ denote its face poset.

(a) Let K, L be convex sets. Prove that

$$\mathcal{F}(K \cap L) = \{F \cap G : F \in \mathcal{F}(K), G \in \mathcal{F}(L), F \cap G \neq \emptyset\}.$$

(b) Let $K \subset \mathbb{R}^m, L \subset \mathbb{R}^n$ be convex sets. Define

$$K \times L = \{(k, l) \in \mathbb{R}^m \times \mathbb{R}^n : k \in K, l \in L\}.$$

Prove that $\mathcal{F}(K \times L) = \mathcal{F}(K) \times \mathcal{F}(L)$.

2. Describe a way to tell if a polyhedron is bounded just by knowing its dimension and the structure of its face poset.
3. Describe all faces of the *hypersimplex*

$$\Delta(n, k) = \text{conv} \left(\sum_{i \in S} e_i : S \subset [n], |S| = k \right) \subset \mathbb{R}^n$$

In addition, give a presentation of $\Delta(n, k)$ by linear equations and inequalities.

4. Fix a positive integer d . For real numbers $t_1 < \cdots < t_n$, let $x(t_i) = (t_i, t_i^2, \dots, t_i^d)$, and let $C = \text{conv}(x(t_i) : i \in [n])$ be a cyclic polytope. Let $S \subset [n]$. Prove that the set $\{x(t_i) : i \in S\}$ is the vertex set of a facet of C if and only if $|S| = d$ and for every $i, j \notin S$, there is an even number of elements of S between i and j .